



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

December 13, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 14576
(TWELFTH REVISION)

EXPIRATION DATE: 2026-10-31

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Structural Composites Industries LLC
Pomona, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale, and use of a non-DOT specification cylinder for the transportation in commerce of the hazardous materials authorized by this special permit. The non-DOT specification cylinders conform to all regulations applicable to fully wrapped carbon-fiber reinforced aluminum lined cylinders (CFFC) except as specified herein. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing special permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 172.101 Hazardous Materials Table Column (9B) in that the quantity limitation for cargo aircraft may not be exceeded and §§ 173.302a(a)(1) and 173.304a(a)(1) in that non-DOT specification cylinders are not authorized, except as specified herein.

5. BASIS: This special permit is based on the application of Structural Composites Industries LLC dated October 28, 2022, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed (containing up to 39% by volume oxygen content)	2.2	UN1002	N/A
Argon, <i>compressed</i>	2.2	UN1006	N/A
Carbon dioxide	2.2	UN1013	N/A
Carbon monoxide, compressed	2.3	UN1016	N/A
Compressed gas, flammable, n.o.s.	2.1	UN1954	N/A
Compressed gas, n.o.s.	2.2	UN1956	N/A
Compressed gas, oxidizing, n.o.s.	2.2	UN3156	N/A
Helium, compressed	2.2	UN1046	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Krypton, compressed	2.2	UN1056	N/A
Liquefied gas, n.o.s.	2.2	UN3163	N/A
Methane, compressed <i>or</i> Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A
Neon, compressed	2.2	UN1065	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Nitrous oxide	2.2	UN1070	N/A
Oxygen, compressed	2.2	UN1072	N/A
Sulfur hexafluoride	2.2	UN1080	N/A
Xenon, compressed	2.2	UN2036	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Prescribed packagings are fully wrapped carbon-fiber reinforced aluminum lined cylinders manufactured in conformance with the Basic Requirements for Fully Wrapped Carbon-Fiber Reinforced Aluminum Lined Cylinders (DOT-CFFC Fifth Revision), dated March 2007, and with the SCI design drawings, titled “Envelope Drawing Wound Pressure Vessel” (Drawing #'s 1275154, 1274655, and 1274986) on file with the Office of Hazardous Materials Safety (OHMS), except as follows:

- (1) **CFFC-2 (Size and Pressure)**: Cylinders made under this special permit are limited to a maximum water volume and service pressure of 315 liters (709 lb) and 517 bar (7,500 psi) respectively, and are subject to the following:
- (2) **CFFC-6(a) (Tensile Tests)**: The tensile specimens may be cut from a sample of representative liner material that has been heat treated in the same heat treatment batch as the all liners for which it represents. Samples of test materials must be of the same material properties as the liner materials it represents.
- (3) **CFFC-10(c) (Drop Tests)**: one cylinder complete with valve must be subjected to a drop test in accordance with ISO 11119-2:2002, section 8.5.8.
- (4) **CFFC-10(g) (Gunfire Test)**: The cylinder (tube) shall be positioned so that the projectile impact point is in the tube sidewall. A 50-caliber gun may be used for thick wall tubes. During the gunfire test, the distance from the firing location to test tube must not exceed 50 yards. If the wall is not penetrated, it is also permissible to fire additional bullets into the same area until the wall is penetrated. Tested tubes shall reveal no evidence of a fragmentation failure. Results of the tests must be recorded.
- (5) **CFFC-10(h) (Bonfire Test)**: Two cylinders must be tested in accordance with all requirements of CFFC-10(h) except the cylinders are in a horizontal position instead of a vertical orientation. The pressurized gas may be vented from other areas of the PRD valve assembly (e.g., melted o-ring) which are affected by intense heat.
- (6) **CFFC-14(iv)**: The REE marking does not apply.

b. Requalification:

- (1) **Hydraulic pressure proof test**: Each cylinder must be requalified once every 5 years by a qualified person holding a valid DOT RIN using a hydraulic proof pressure test equal to 1.5 times the marked service pressure. The pressure must be held for a minimum of 3 minutes without a loss of pressure. The hydraulic pressure proof test may be substituted by a pneumatic pressure proof test subject to the following conditions:

(i) The testing facility is prepared and approved for pressure proof test based on a documented risk assessment and failure mode and effect analysis.

(ii) Each cylinder is pressurized to 1.25 times the marked service pressure. The pressure must be held for 10 consecutive minutes. The loss of pressure during the hold time must not exceed 5% of the original test pressure.

(iii) In case of equipment failure during the test which results in loss of pressure prior to the completion of the 10-minute hold time, the test may only be repeated one time, at a pressure of 1.27 times the marked service pressure.

(iv) A cylinder that fails to maintain the test pressure must be evaluated to determine the cause. A cylinder that shows evidence of damage (surface distortion, unraveled fibers or other evidence of weakness) must be evaluated in accordance with the visual inspection procedure described in this special permit.

(2) Visual Inspection: Each cylinder must be visually inspected in accordance with CGA Pamphlet C-6.2 Guidelines for Visual Inspection and Re-qualification of Fiber Reinforced High Pressure Cylinders”, except as specifically noted herein:

(i) Cylinders with fiber damage (cuts, abrasions, etc.) that exceeds Level 1 type damage as defined in CGA Pamphlet C-6.2 and meet the following depth and length criteria are considered to have Level 2 damage:

(A) Depth: Damage that upon visual inspection is seen to penetrate the outer fiberglass layer but does not expose the carbon layer beneath, or that has a measured depth of greater than 0.005 inch and less than 0.045 inch for cylinders with an outside diameter greater than 7.5 inches or less than 0.035 inch for cylinders 7.5 inches or less in outside diameter.

(B) Length: Damage that has a maximum allowable length of:

Region	Direction of fiber damage	Maximum length of damage
Cylinder sidewall and domes	Transverse to fiber direction (longitudinal direction)	20% of the straight sidewall section length
Cylinder sidewall and domes	In fiber direction (circumferential direction)	20% of the straight sidewall section length

(ii) Cylinders with damage that meet the Level 2 criteria must be rejected. Retesters must contact the cylinder manufacturer in the event that the damage cannot be clearly interpreted based on these criteria. Repair of rejected cylinders is authorized for Level 2 type damage. Repairs must be made in accordance with CGA Pamphlet C-6.2, prior to the hydrostatic pressure test. Repairs must be evaluated after the hydrostatic test.

(iii) Cylinders that have direct fiber damage that penetrates through the outer fiberglass layer and into the carbon layer, or that have a measured damage depth of greater than the Level 2 maximum are considered to have Level 3 type damage. Cylinders that have damage with depth meeting Level 2, but length exceeding the Level 2 maximum are considered to have Level 3 type damage. Cylinders with Level 3 type damage are not authorized to be repaired and must be condemned.

(iv) A hydrostatic requalification may be repeated as provided in § 180.205(g); only two such tests are permitted. Pressurization prior to the official hydrostatic test for the purpose of a systems check may not exceed 85% of the minimum required test pressure.

(3) Persons who perform inspection and testing of cylinders subject to this special permit must comply with § 180.205(b) and with all the terms and conditions of this special permit.

(4) The requalification date (month/year) must be permanently marked on the cylinder as specified in CFFC-14. The marking of the RIN symbol on the cylinder certifies compliance with all the terms and conditions of this special permit.

c. OPERATIONAL CONTROLS:

(1) A cylinder manufactured under this special permit must be removed from service after 15 years from the date of manufacture.

- (2) Cylinders filled with mixtures of carbon monoxide and hydrogen must not contain a moisture content in excess of 50 parts per million.
- (3) The maximum quantity authorized for cylinders containing mixtures of carbon monoxide and hydrogen is 25 kilograms (kg).
- (4) A cylinder that has been subjected to fire may not be returned to service.
- (5) Cylinders used in oxygen service must conform with § 173.302a(a)(5)(i) through (iv).
- (6) Cylinders used in nitrous oxide service must conform with § 173.304a(a)(1).
- (7) Cylinders must be manifolded in accordance with the requirements of § 173.301(g).
- (8) Transportation of Division 2.1 (flammable gas) and Division 2.3 (gas poisonous by inhalation) hazardous materials is not authorized aboard cargo vessel and aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).
- (9) Transportation of oxygen is only authorized aboard aircraft when in accordance with § 175.501.
- (10) Transportation of compressed oxygen and oxidizing gases specified in paragraph 6 of this special permit aboard cargo only aircraft must comply with §§ 173.302(f) and 173.304(f), as applicable.
- (11) Cylinders may not be used for underwater breathing purposes.
- (12) The cylinders are permanently mounted within a structural frame during transportation. Structural framework that is intended for transportation of the cylinders under this special permit must have an appropriate engineering calculation (e.g., Finite Element Analysis (FEA)). The report must be submitted to OHMS. The calculation must demonstrate the framework's ability to protect the cylinders from catastrophic damage (rupture) due to front, rear, or side impact, and rollover. As a minimum, the frame must be designed to meet the following:
 - (i) All requirements of § 173.301(i).
 - (ii) The frame design must meet all requirements of CGA TB-25.
- (13) All cylinders must be operated and maintained in accordance with Structural Composites Industries Cylinder Component Operations Manual.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a packaging covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit and must be made available to a DOT representative upon request.

e. Each packaging manufactured under the authority of this special permit must be either: (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured; or (2) marked with a registration symbol designated for a specific manufacturing facility by OHMS for a specific manufacturing facility.

f. The cylinders described in this special permit are authorized only for normal transportation as an article of commerce i.e., the movement of hazardous materials packages from consignor to consignee.

g. When authorized for transportation by cargo vessel as prescribed in the § 172.101 Hazardous Materials Table, flammable gases covered by this special permit must be packed within a closed freight container of steel construction.

h. Each cylinder must be plainly and durably marked "DOT-SP 14576" as specified in §§ 172.302(b) and (c). Additionally, each framework must be marked "DOT-SP 14576".

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, and cargo-only aircraft.10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft, or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

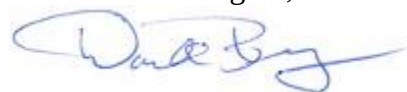
Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) – “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 Immediate notice of certain hazardous materials incidents, and 171.16 Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BrMoore/NICKS